Modern Physics — Lectures 4, 5 & 6

Study guide: de Broglie waves & uncertainty → wavefunctions & electron spin → Pauli, spin–orbit & X-rays

This is a plain-English rewrite of your three slide decks. Lectures 5 and 6 overlap (the Lecture 6 deck repeats the Stern–Gerlach and spin slides), so I merged them into one storyline. Every section follows the same pattern: **the question → the idea → the maths → what to remember**. Anything I added that is *not* on the slides is marked in green, and slide typos I found are collected on the last page. (The orbital quantum number is written ℓ here so it does not look like a 1 or an I.)

The whole story in 8 steps

#	What happens	Where
1	Single photons, sent one at a time, still build up a wave diffraction pattern. The pattern tells you the **probability** of where a photon lands.	Lec 4
2	Electrons also diffract. So every particle has a wavelength: **$\lambda = h/p$**.	Lec 4
3	Squeeze a particle through a slit: you learn its position better, but its sideways momentum spreads out. That is the **uncertainty principle**.	Lec 4
4	So exact trajectories are out. Replace them with a **wavefunction ψ**: $\lvert\psi\rvert^2$ = probability. It obeys the **Schrödinger equation**.	Lec 5
5	Solving hydrogen gives quantum numbers **n, ℓ, m_ℓ**. A magnetic field splits the m_ℓ levels (Zeeman); Stern–Gerlach shows this as spots on a screen.	Lec 5
6	But $\ell = 0$ atoms give **two** spots, not one. Conclusion: the electron has an intrinsic **spin** ($s = ½$).	Lec 5–6
7	Spin ½ makes electrons **fermions** → **Pauli exclusion**. Also L and S interact (**spin–orbit**), splitting levels slightly (**fine structure**).	Lec 6
8	Knock out an inner electron, and an outer one falls in, emitting an **X-ray** with energy characteristic of the element. Fine structure splits Kα into $K\alpha_1$ and $K\alpha_2$.	Lec 6

Symbols and constants you will meet

Symbol	Meaning	Symbol	Meaning
h	Planck's constant, 6.63×10^{-34} J·s	ψ	wavefunction (the quantum state)
$\hbar$	$h/2\pi \approx 1.055\times10^{-34}$ J·s	$\hat{H}$	Hamiltonian: the total-energy operator
λ, ν	wavelength, frequency	n, ℓ, m_ℓ	orbital quantum numbers (energy, angular-momentum size, its z-part)
p, E	momentum, energy	s, m_s	spin quantum numbers (electron: ½ and ±½)
m_e, e	electron mass 9.11×10^{-31} kg, charge 1.602×10^{-19} C	L, S, J	orbital, spin and total angular momentum
μ, B	magnetic moment, magnetic field	Z	atomic number (protons in the nucleus)
~	"of the order of" (rough size)	$\gtrsim$, $\ll$	"greater than or about", "much smaller than"

Part 1 · Lecture 4: de Broglie waves & the uncertainty principle

1.1 Photons, one at a time, still make a wave pattern

Shine light through a single narrow slit and wave optics predicts a bright central band with dimmer side fringes. Now turn the intensity down so that **one photon at a time** goes through (all with the same energy). Each photon makes exactly **one dot** on the screen.

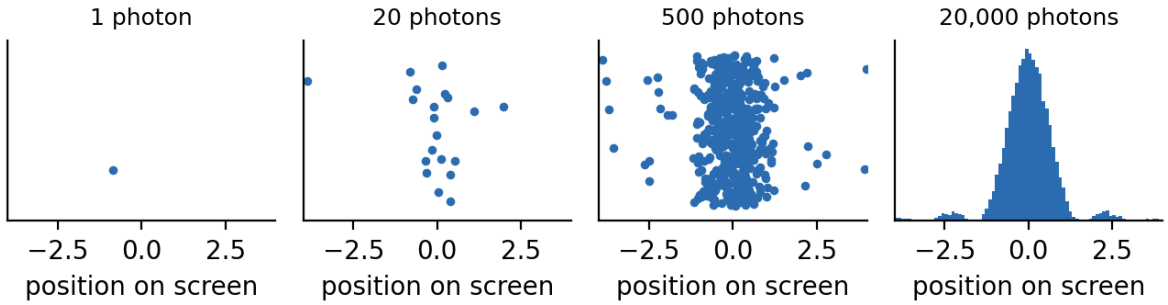


Fig 1. Dots accumulating. No single dot looks like a wave, but thousands together reproduce the diffraction pattern exactly.

- You **cannot predict where the next photon lands**, and positions are not equally likely.
- You **can predict the distribution** after many photons. The wave description did not vanish; it now gives the **statistics**.
- Wave picture: the y-axis is *intensity*. Particle picture: the y-axis is *number of photons N(y)*. They match, because brighter = more photons.
- For a single photon, "more photons at this spot" has to mean "**higher probability of the hit being here**".

> **Key idea**
> The diffraction pattern is the **probability distribution** for where one particle will be detected.

1.2 Wave–particle duality

You might think photons diffract only because they come from an electromagnetic wave. **No**: electrons (and protons, neutrons) also produce diffraction and interference, and there is no classical "electron wave" to start from. **Wave-particle duality** means quantum objects show behaviour we usually associate with *either* particles *or* waves.

Two questions drive the rest of the lecture: **(Q1)** what does the *setup* (the slit) tell us about the particle? **(Q2)** what does the *diffraction pattern* tell us about the particle?

1.3 Q1: what is the particle's "wavelength"? (de Broglie)

Diffraction is clearly visible only when the wavelength is comparable to the slit width: **λ ~ a**. For λ ≪ a the spreading is tiny (a narrow blob). For λ > a you just get one smeared central band. The first dark fringes of a single slit sit at

$$a \sin\theta = m\,\lambda \quad (m = \pm 1, \pm 2, \ldots)$$

A particle going through the same slit needs a λ too. Start from what we know about the photon, which has E = hν and (from relativity, since it travels at c, its rest mass is zero) E = pc:

$$pc = h\nu = hc/\lambda \Longrightarrow \boldsymbol{\lambda = h / p}$$

De Broglie's proposal: **define λ = h/p for any particle**. The slides stress this is not a claim that the particle physically wiggles up and down. It is a **length scale**: compare it with the size of your experiment. If λ is comparable to the apertures or spacings involved, wave effects (diffraction, interference) matter. If λ is far smaller, the object behaves classically.

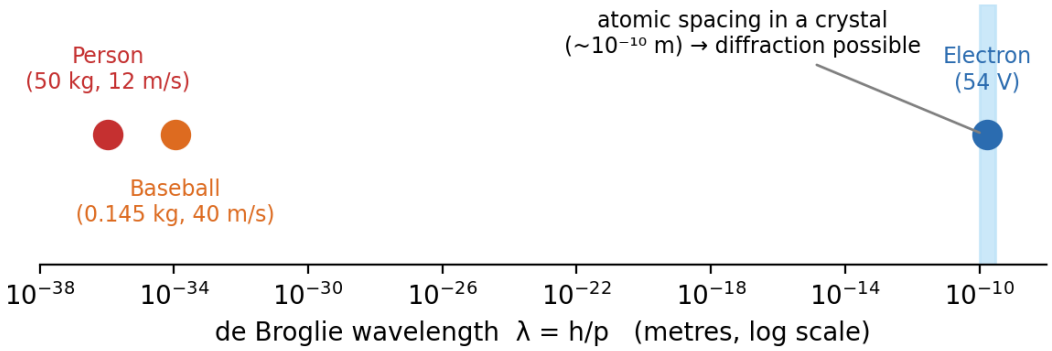


Fig 2. Why you do not diffract through a doorway but an electron does through a crystal.

$$\text{Person: } \lambda = h/(mv) = 6.63\times10^{-34} / (50 \times 12) \approx 1.1\times10^{-36} \text{ m}$$

To diffract a 50 kg person running at 12 m/s you would need a slit of about 10^{-36} m, absurdly smaller than anything physical. Hence we never see quantum behaviour in everyday life.

> **Key idea**
> λ = h/p. The momentum decides the length scale. Large momentum → tiny λ → classical.
> Compare λ with the size of the experiment (λ ≪ size: classical, λ ~ size: quantum).

1.4 Electron diffraction (a real, measurable case)

Accelerate an electron through voltage V. Its kinetic energy is $K = eV = p^2/2m_e$, so $p = \sqrt{(2m_e eV)}$ and

$$\lambda = h / \sqrt{(2\, m_e\, e\, V)} \approx 1.227 / \sqrt{V} \text{ nm (V in volts)}$$

For V = 54 V: λ = 1.227/√54 ≈ **0.167 nm**. That is about the spacing between atoms in a nickel crystal (d = 0.091 nm for Ni(111)), so the crystal acts as a diffraction grating. The condition for constructive interference of waves bouncing off crystal planes is **Bragg's law**:

$$n\,\lambda = 2\,d\,\sin\theta$$

With n = 1: sin θ = 0.167/(2×0.091) ≈ 0.92, so θ ≈ 67° from the crystal plane. The slide shows a measured scattering peak around 50° for 54 V, and the intensity changes with voltage (68 V, 92 V give different λ), which is how the wave nature is confirmed. Your Physics Lab has this setup.

1.5 Q2: what does the diffraction pattern tell us? (Uncertainty principle)

Before the slit, every photon moves purely along x, so $p_y = 0$. After the slit, photons land away from the centre, so they must have **gained some y-momentum**. Where did it come from? From the confinement by the slit itself. Here is the derivation:

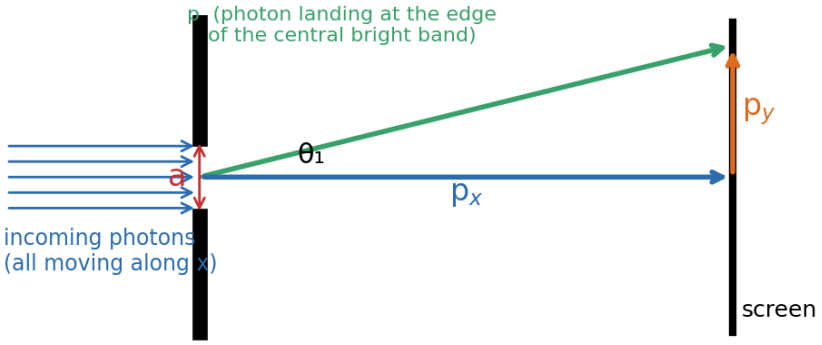


Fig 3. A photon landing at the edge of the central maximum has momentum components p_x and p_y.

- **Step 1.** Split the momentum: $p_x = p \cos \theta_1$, $p_y = p \sin \theta_1$, so $p_y/p_x = \tan \theta_1$.
- **Step 2.** For small angles $\tan \theta_1 \approx \theta_1$, and the first minimum gives $\theta_1 \approx \lambda/a$. So $\mathbf{p_y \approx p_x \lambda / a}$.
- **Step 3.** Photons in the central band therefore have p_y anywhere between $-p_x\lambda/a$ and $+p_x\lambda/a$. That spread is the uncertainty: $\Delta p_y \gtrsim p_x\lambda/a$.
- **Step 4.** Use de Broglie, $\lambda = h/p_x$: $\Delta p_y \gtrsim (p_x/a)(h/p_x) = \mathbf{h/a}$.
- **Step 5.** The slit tells us where the photon passed: $\Delta y \sim a$. Multiply:

$$\Delta y \cdot \Delta p_y \sim h \text{ (precise statement: } \Delta y \cdot \Delta p_y \geq \hbar/2 = h/4\pi)$$

Slit width a	Position knowledge Δy ~ a	Diffraction pattern	Momentum spread Δp_y
wider ↑	worse (larger Δy)	narrower	smaller
narrower ↓	better (smaller Δy)	wider	larger

Key idea

Better localisation in position ⟹ larger spread in momentum.
This is not clumsy measurement. It is built into how quantum objects work, and it comes straight from the wave behaviour.

Part 2 · Lecture 5: quantum mechanics and electron spin

2.1 Which description should I use?

The slides open with two everyday examples. A phone receiving a signal is explained by the **electromagnetic wave** (the oscillating field drives charges in the antenna). A solar calculator is explained by **photon–electron interaction** (a photon lifts an electron across a semiconductor band gap). The physics did not change between the two; the **relevant scale** did. Before modelling anything, ask: what are the characteristic length and energy scales, and how do they compare with λ_{dB}? If $\mathbf{L \sim \lambda_{dB}}$, quantum effects can matter.

2.2 Why trajectories fail

In classical mechanics, knowing r(0) and p(0) lets Newton's equation give the whole path r(t). In our diffraction experiment we prepare identical particles in the identical way and still get different detection positions. And $\Delta y\, \Delta p_y \geq \hbar/2$ says exact position *and* exact momentum cannot both exist. Two independent clues, same conclusion:

Clue	What it says
Diffraction	Outcomes are **probabilistic**
Uncertainty principle	There is **no exact (y, p_y)** pair to build a trajectory from

2.3 The wavefunction and the Schrödinger equation

What replaces the trajectory is the **wavefunction ψ(r, t)**, which describes the quantum state. It does not hide a secret path; it contains what you need to compute the probabilities of measurement outcomes:

$$|\psi(r, t)|^2 = \text{probability density of finding the particle at r}$$

$$|\psi(y)|^2\, dy = \text{probability of detection between y and y + dy}$$

Newton's law F = ma had an equation of motion. For ψ it is the **Schrödinger equation**:

$$i\hbar\, \partial\psi/\partial t = \hat{H}\, \psi$$

Here $\hat{H}$ is the **Hamiltonian operator**, associated with the total energy. For states of definite energy this reduces to the **time-independent Schrödinger equation $\hat{H}\psi = E\psi$**, whose solutions give the allowed states ψ_n and energies E_n.

2.4 Observables become operators

Heisenberg built the theory around measurable quantities (atomic spectra show discrete energy jumps). The maths that came out did not obey ordinary multiplication: the order mattered, **AB ≠ BA**, exactly like matrices. So in QM each measurable quantity becomes an **operator**: x → $\hat{x}$, p → $\hat{p}$, E → $\hat{H}$. An operator needs something to act on, just as a matrix acts on a vector: $\hat{H}\psi$ means "apply the energy operator to the state".

2.5 Classical recipe vs quantum recipe

	Classical	Quantum
1. Specify the system	Force law / potential, e.g. $V(r) = -GMm/r$	Potential, e.g. $V = 0$ or $V = -e^2/(4\pi\varepsilon_0 r)$
2. Solve	Newton: $F = -\nabla V = ma$	$\hat{H}\psi = E\psi$
3. What you get	Which trajectories are possible (circle, ellipse…)	Allowed states and quantities: ψ_n, E_n, L^2, L_z…
4. Initial condition	$r(0)$, $p(0)$ pick one trajectory	$\psi(r, 0)$ fixes the evolution $\psi(r, t)$ and the probabilities $\lvert\psi\rvert^2$

The slides recap Prof. Vandana's particle-in-a-1D-box example as the model for this recipe: $V = 0$ inside $0 < x < L$, infinite outside, so $\psi(0) = \psi(L) = 0$. This gives $\psi_n = \sqrt{(2/L)}\sin(n\pi x/L)$ and $E_n = n^2\pi^2\hbar^2/(2mL^2)$, $n = 1, 2, 3, \ldots$ Only discrete energies survive the boundary conditions.

2.6 The hydrogen atom and its three quantum numbers

The electron feels the Coulomb potential $V(r) = -e^2/(4\pi\varepsilon_0 r)$. Solving $\hat{H}\psi = E\psi$ gives wavefunctions $\psi_{n\,\ell\,m_\ell}(r, \theta, \varphi)$ and energies $E_n = -13.6/n^2$ eV. Two more quantised quantities come out automatically because V depends only on $|r|$ (spherical symmetry, unchanged by rotations), so angular momentum is conserved:

Quantum number	Allowed values	What it fixes
n	$1, 2, 3, \ldots$	Energy: $E_n = -13.6/n^2$ eV
ℓ	$0, 1, \ldots, n-1$	Size of orbital angular momentum: $L^2 = \ell(\ell+1)\hbar^2$
m_ℓ	$-\ell, -\ell+1, \ldots, 0, \ldots, \ell$	Component along a chosen axis: $L_z = m_\ell\hbar$ ($2\ell+1$ values)

With no external field the energy depends only on n, so all (ℓ, m_ℓ) states of the same n are **degenerate**. Small corrections (relativistic kinetic energy, spin–orbit coupling, the Darwin term) form $H_{extra} \ll H$ and lift part of this degeneracy: that is **fine structure** (we return to it in Part 3).

2.7 Orbital magnetic moment and the Zeeman effect

An electron orbiting the nucleus is a tiny current loop, and a current loop is a magnetic dipole. Period $T = 2\pi R/v$, current $I = -e/T = -ev/(2\pi R)$, area $A = \pi R^2$:

$$\mu = I\,A = -(e/2)\,R\,v\,,\; L = m_e\,R\,v \Longrightarrow \boldsymbol{\mu_L = -(e\,/\,2m_e)\,L}$$

Put the atom in a **uniform** field $B = B\hat{z}$. The interaction energy is $U = -\mu_L \cdot B = (e/2m_e)\,L_z B = (e\hbar/2m_e)\,m_\ell\,B$, i.e. $\mathbf{U = \mu_B\,m_\ell\,B}$ with the Bohr magneton $\mu_B = e\hbar/2m_e$. The $(2\ell+1)$ degenerate levels split evenly, one for each m_ℓ. This is the **Zeeman effect**: a splitting of *energy levels*.

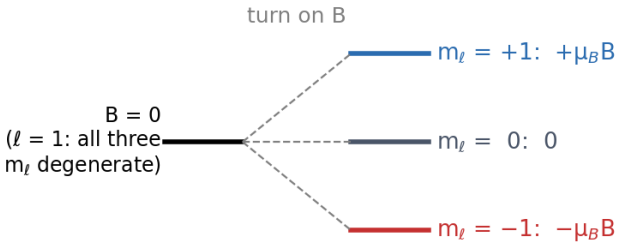


Fig 4. Zeeman splitting for $\ell = 1$ (a p state).

In a uniform field the net force on the dipole is zero (the atom is not pushed anywhere), only a torque acts.

2.8 Stern–Gerlach: seeing m_ℓ as spots on a screen

Now use a **non-uniform** field with dB/dz ≠ 0. The energy is $U = -\mu_z B(z)$, so the atom feels a **force** $F_z = \mu_z\, dB/dz$. Since $\mu_z = -(e\hbar/2m_e)\, m_\ell$, the force depends on m_ℓ:

$$F_z = -(e\hbar/2m_e)\, m_\ell\, dB/dz$$

A beam of neutral atoms (e.g. silver) from a hot oven passes through collimating slits and the magnet. Each atom is deflected up, down, or not at all depending on its m_ℓ, and lands on a detector screen. **Prediction for $\ell = 1$:** three spots ($m_\ell = +1, 0, -1$). **Prediction for $\ell = 0$:** $m_\ell = 0$ only, so no force and a single spot at the centre.

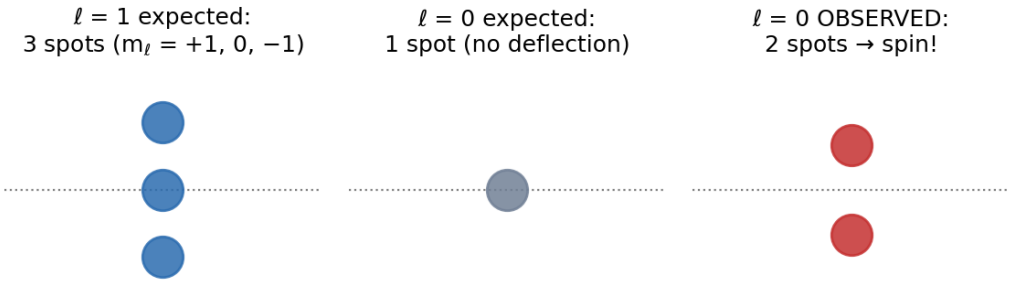


Fig 5. The surprise: $\ell = 0$ atoms (hydrogen ground state) split into two spots.

> **Watch out**
>
> **Zeeman ≠ Stern–Gerlach.** Zeeman: levels split *in energy* (uniform field). Stern–Gerlach: atoms split *in space* (non-uniform field, force). Same cause (m_ℓ and the magnetic moment), different observable.

2.9 Electron spin

For $\ell = 0$ there is no orbital angular momentum, so the force from Section 2.8 should be zero. Yet we see two spots, i.e. two deflection directions, so the atom must carry **another** source of magnetic moment. The electron has an **intrinsic angular momentum S, called spin**:

$$s = ½,\ m_s = \pm ½,\ S^2 = s(s+1)\hbar^2 = ¾\hbar^2,\ S_z = m_s\hbar$$

Two allowed m_s values means two spots. The general rule is the same as for L: m_s runs from $-s$ to $+s$ in integer steps. (**Note:** the slide writes "s = 0, 1, 2, …, n−1", which is copy-pasted from the ℓ rule. For an electron s is simply fixed at ½.)

Is spin a little ball rotating? The slides test it. If the electron were a sphere of radius R ~ 10^{-15} m with S ~ m_eRv ~ ħ/2, the surface speed would be v ~ ħ/(2m_eR) ~ 10^{11} m/s, **far faster than light**. So no: spin is an intrinsic quantum property with no classical rotating-object picture.

> **Key idea**
>
> $\psi_{n\,\ell\,m_\ell}$ alone is **not** the complete description of an electron. You also need the spin state ($m_s = \pm ½$). An electron in an atom now has **two** angular momenta, L and S. Does the energy depend on their relative orientation? Lecture 6 says yes.

Part 3 · Lecture 6: Pauli exclusion, spin–orbit coupling & X-ray spectra

3.1 Fermions and bosons

Spin is intrinsic: it is a fixed property of each kind of particle. Electrons have s = ½, photons s = 1, the Higgs s = 0. Particles fall into two families:

Family	Spin	Examples	Wavefunction under exchange
Bosons	integer: 0, 1, 2, …	photon (1), Higgs (0)	symmetric
Fermions	half-integer: ½, 3/2, …	electron, proton, neutron (½)	antisymmetric → Pauli exclusion

3.2 Pauli exclusion principle

Statement: two identical electrons cannot occupy the same complete quantum state. What counts as "the state" depends on the setting:

Setting	State labelled by	Not allowed	Allowed
Free electron	(p, m_s)	$(p_1, +½)$ and $(p_1, +½)$	$(p_1, +½)$ and $(p_1, -½)$; or $(p_1, +½)$ and $(p_2, +½)$
Electron in hydrogen	(n, ℓ, m_ℓ, m_s)	identical in all four numbers	opposite m_s, or different spatial state (n', ℓ', m_ℓ')
1s orbital $(1, 0, 0, m_s)$	—	(1,0,0,+½) twice	(1,0,0,+½) with (1,0,0,−½): so 1s holds at most 2 electrons

Why is it true? The exchange argument. Electrons are identical, so we cannot say which is "1" and which is "2". The two-electron wavefunction must therefore contain both assignments of states A and B:

$$\Psi(1,2) = (1/\sqrt{2})\,[\,\psi_A(1)\psi_B(2) \pm \psi_A(2)\psi_B(1)\,]$$

Which sign? Put both electrons in the *same* state (B = A):

- **+ sign:** $\Psi = \sqrt{2}\,\psi_A(1)\psi_A(2) \neq 0$. Both in the same state is allowed. (This is what bosons do.)
- **− sign:** $\Psi = (1/\sqrt{2})[\psi_A(1)\psi_A(2) - \psi_A(2)\psi_A(1)] = \mathbf{0}$. Impossible. The state cannot exist.

> **Key idea**
> Electrons obey Pauli, so they take the − sign: **$\Psi(1,2) = -\Psi(2,1)$** (antisymmetric under exchange). Fermions → antisymmetric wavefunction → Pauli exclusion. Bosons → symmetric wavefunction (no restriction).

A subtle slide (the wave-packet one). Two electrons with exactly the same momentum look identical in momentum. But exact momentum means $\Delta p = 0$, so by uncertainty $\Delta x \to \infty$: the electron is spread everywhere and you cannot assign them different locations. Using localised **wave packets** centred at well-separated positions x_1 and x_2 (same average momentum) gives genuinely different states, so both are allowed.

Extra (not on the slides)

Pauli exclusion is why atoms have **shells**: electrons cannot all drop into 1s, so they fill 1s, then 2s, 2p, … This underlies the periodic table and why matter takes up space.

3.3 Spin–orbit interaction

Lecture 5 ended by asking: *can we expect a Zeeman-like effect from spin?* Yes. In an external field B the spin energy is $\Delta H = -\mu_S \cdot B$ with $\mu_S = -g_s(e/2m_e)\,S$ and $g_s \approx 2$, giving two levels $m_s = \pm ½$. But the more important point is that **even with $B_{ext} = 0$ the electron's spin feels a magnetic field**. The argument, step by step:

- In the **nucleus frame** the electron orbits in the nucleus's electric field.
- In the **electron's rest frame** the positively charged nucleus appears to circle around it.
- A circulating positive charge is a current loop, so it produces a magnetic field $\mathbf{B_{eff}}$, perpendicular to the orbital plane, i.e. $\mathbf{B_{eff} \parallel L}$. The slides give $B_{eff} = (\mu_0/4\pi)(Ze/m_e r^3)\,L$.
- The electron's spin moment $\mu_S \propto S$ sits in that field, so $U = -\mu_S \cdot B_{eff} \propto \mathbf{L \cdot S}$.

$$\hat{H} = \hat{H}_0 + \hat{H}_{SO},\ \hat{H}_{SO} = \xi(r)\ L\cdot S,\ L\cdot S = LS\cos\theta$$

If L and S are more **aligned**, $L\cdot S > 0$ and the energy goes **up**. If more **opposed**, $L\cdot S < 0$ and the energy goes **down**. So the energy depends on the relative orientation of the orbit and the spin.

Using total angular momentum J = L + S. Since L and S interact, neither is separately fixed; J is. Squaring $J = L + S$ gives $J^2 = L^2 + S^2 + 2\,L\cdot S$, so

$$L\cdot S = ½\,(\,J^2 - L^2 - S^2\,) = (\hbar^2/2)\,[\,j(j+1) - \ell(\ell+1) - s(s+1)\,]$$

with $j = |\ell - s|, \ldots, \ell + s$. For the **2p** level ($\ell = 1$, $s = ½$):

Level	j	Number of states	L·S	Energy
$2p_{3/2}$	3/2	4 ($m_j = \pm 3/2, \pm 1/2$)	$+\hbar^2/2$	higher
$2p_{1/2}$	1/2	2 ($m_j = \pm 1/2$)	$-\hbar^2$	lower

(Check: $4 + 2 = 6 = 2(2\ell+1)$ states, nothing lost.) So **2p is not one level**: it splits into $2p_{1/2}$ and $2p_{3/2}$. An s level ($\ell = 0$) has only $j = ½$, so it does not split. The notation is $n\,\ell_j$. This tiny splitting is the **fine structure**: for n = 2, $2p_{3/2}$ and $2p_{1/2}$ separate from each other, while $2s_{1/2}$ stays single; for n = 3 you get $3p_{3/2}$, $3p_{1/2}$, $3d_{5/2}$, $3d_{3/2}$ and $3s_{1/2}$.

3.4 X-ray spectra

How characteristic X-rays arise.

- An energetic electron hits an atom and knocks out an **inner-shell** electron. The ejected electron is free; the atom is left excited with a **vacancy**.
- An electron from a higher shell drops into the vacancy and the atom emits a photon with $\mathbf{E_\gamma = E_i - E_f}$.
- Naming: **L → K** gives **Kα**; **M → K** gives **Kβ**. Because the energies are set by the atom's own levels, the lines are **characteristic of the element**.

Is every L → K jump an X-ray? No. The photon energy is just the level difference; whether it lands in the X-ray range depends on Z. For a hydrogen-like atom $E_n = -13.6\,Z^2/n^2$ eV, so the energies grow like Z^2. In many-electron atoms the other electrons **screen** the nucleus, so effectively $Z \to Z_{eff} < Z$.

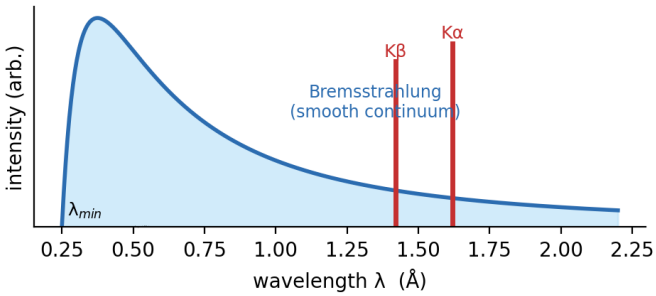


Fig 6. X-ray tube spectrum (schematic): smooth continuum plus sharp characteristic lines.

An X-ray tube (electrons accelerated onto a metal anode) uses **two mechanisms** at once:

Mechanism	Spectrum shape
Electrons decelerating in the anode (**Bremsstrahlung**, “braking radiation”)	Smooth continuum, with a minimum wavelength λ_{min} (the slide’s plot shows 0.25 Å)
Inner-shell transitions in the anode atoms	Sharp **characteristic lines** (Kα, Kβ, …)

Fine structure shows up in X-rays. Since 2p is really $2p_{3/2}$ and $2p_{1/2}$ (Section 3.3), the L → K transition has **two** versions, both ending on $1s_{1/2}$ (which does not split):

$$2p_{3/2} \rightarrow 1s_{1/2} \Longrightarrow \mathbf{K\alpha_1}$$

$$2p_{1/2} \rightarrow 1s_{1/2} \Longrightarrow \mathbf{K\alpha_2}$$

Since $E(2p_{3/2}) \neq E(2p_{1/2})$, the two photons have slightly different energies: $E_{K\alpha 1} \neq E_{K\alpha 2}$.

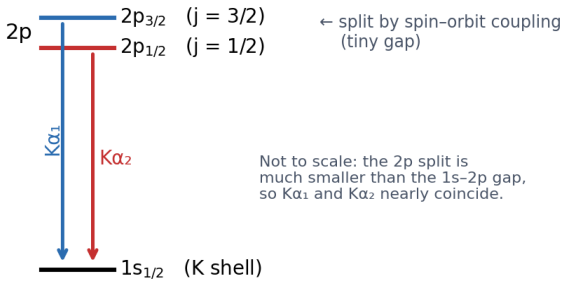


Fig 7. Kα is really a close doublet.

Extra (not on the slides)

Moseley’s law (rough): $E(K\alpha) \approx (3/4)(13.6\text{ eV})(Z-1)^2$. For copper, Z = 29: $0.75 \times 13.6 \times 28^2 \approx 8.0$ keV. Measured: about 8.05 keV. The (Z−1) is the screening correction ($Z_{eff} \approx Z-1$ for the K shell).
The minimum wavelength of the continuum is set by the tube voltage: $\lambda_{min} = hc/(eV)$ (all of one electron’s energy into one photon).

Formula cheat sheet

Topic	Formula	Notes
Photon	$E = h\nu = pc = hc/\lambda$	rest mass 0
de Broglie	$\lambda = h/p$	any object; compare with the experiment's length scale
Electron through V	$\lambda = h/\sqrt{(2m_e eV)} \approx 1.227/\sqrt{V}$ nm	non-relativistic
Single slit minima	$a \sin\theta = m\lambda$, $m = \pm1, \pm2, \ldots$	$\theta_1 \approx \lambda/a$
Bragg	$n\lambda = 2d \sin\theta$	crystal planes, spacing d
Uncertainty	$\Delta y\, \Delta p_y \geq \hbar/2$ (≈ h order of magnitude)	$\hbar = h/2\pi$
Probability	$\lvert\psi(r,t)\rvert^2$ = density; $\lvert\psi(y)\rvert^2 dy$ = P(y to y+dy)	ψ is the state
Schrödinger	$i\hbar\, \partial\psi/\partial t = \hat{H}\psi$; $\hat{H}\psi = E\psi$ (stationary)	$\hat{H}$: total-energy operator
Hydrogen energy	$E_n = -13.6/n^2$ eV	n = 1, 2, …
Orbital L	$L^2 = \ell(\ell+1)\hbar^2$; $L_z = m_\ell \hbar$	$\ell = 0 \ldots n-1$; $m_\ell = -\ell \ldots \ell$
Orbital moment	$\mu_L = -(e/2m_e)\, L$	$\mu_B = e\hbar/2m_e$
Zeeman (orbital)	$\Delta E = \mu_B\, m_\ell\, B$	energy-level splitting
Stern–Gerlach force	$F_z = \mu_z\, dB/dz = -(e\hbar/2m_e)\, m_\ell\, dB/dz$	spatial splitting on screen
Spin	$s = ½$; $S^2 = ¾\hbar^2$; $S_z = \pm\hbar/2$	intrinsic; not a literal spin
Pauli	$\Psi(1,2) = -\Psi(2,1)$ (fermions)	no two electrons in the same (n,ℓ,m_ℓ,m_s)
Spin–orbit	$\hat{H}_{SO} = \xi(r)\, L\cdot S$; $L\cdot S = (\hbar^2/2)[j(j+1) - \ell(\ell+1) - s(s+1)]$	$J = L + S$, $j = \lvert\ell - s\rvert \ldots \ell + s$
X-ray photon	$E_\gamma = E_i - E_f$; hydrogen-like $E_n = -13.6\, Z^2/n^2$ eV	$Z \to Z_{eff}$ in real atoms

Common confusions, answered

- **"Is the particle physically oscillating at its de Broglie wavelength?"** No. λ = h/p is a length scale telling you when wave effects matter.
- **"Is uncertainty just a limit of our instruments?"** No. It follows from wave behaviour: a narrower slit is a sharper position and a wider momentum spread.
- **"Does the theory hide a real trajectory?"** The slides say no: it evolves a quantum state and gives probabilities of outcomes.
- **"Zeeman or Stern–Gerlach?"** Uniform field → energy splitting. Non-uniform field → force → spatial splitting.
- **"Is spin the electron rotating?"** No, that would need speeds above c. Spin is intrinsic.
- **"Why can't two electrons share a state?"** Their combined wavefunction must flip sign when they are swapped. Putting both in one state would force Ψ = 0.

Self-test (answers below)

- **Q1.** Compute the de Broglie wavelength of an electron accelerated through 100 V.
- **Q2.** An electron is confined to a region of width 0.1 nm (atom-sized). Estimate the minimum Δp.
- **Q3.** You halve the slit width. What happens to the central diffraction band and to Δp_y?
- **Q4.** A hydrogen atom is in a state with n = 3. List the possible ℓ values, and how many m_ℓ values each has.
- **Q5.** Why do we see two Stern–Gerlach spots for hydrogen in its $\ell = 0$ ground state?
- **Q6.** How many distinct (n, ℓ, m_ℓ, m_s) states exist for n = 2? What does Pauli say about the electrons that fill them?
- **Q7.** For the 2p level, what values can j take, and which of $2p_{3/2}$ and $2p_{1/2}$ has higher energy?
- **Q8.** Why are $K\alpha_1$ and $K\alpha_2$ slightly different in energy?

Answers

A1. $\lambda = 1.227/\sqrt{100} = 0.1227$ nm (≈ 0.123 nm). **A2.** $\Delta p \geq \hbar/(2\Delta x) = 1.055\times10^{-34}/(2\times10^{-10}) \approx 5\times10^{-25}$ kg·m/s. **A3.** The pattern gets **wider** ($\theta_1 \approx \lambda/a$ doubles) and $\Delta p_y \gtrsim h/a$ **doubles**, because we now know the position better. **A4.** $\ell = 0$ (1 value of m_ℓ), $\ell = 1$ (3), $\ell = 2$ (5): 9 orbital states in total (n^2). **A5.** The electron has intrinsic spin ½ with $m_s = \pm$½, so there are two magnetic-moment orientations, even though L = 0. **A6.** $2n^2 = 8$ states (2s: 2, 2p: 6). Pauli lets each be occupied by at most one electron. **A7.** j = 3/2 or 1/2; $2p_{3/2}$ is higher (L·S > 0 when L and S are more aligned). **A8.** They start from different fine-structure levels ($2p_{3/2}$ vs $2p_{1/2}$) and end on the same $1s_{1/2}$.

Typos and loose ends in the slides

- **Lecture 4, “Momentum of photons in central maxima”:** the range is printed as $-p_x\lambda/a \lesssim p_y \lesssim -p_x\lambda/a$. The upper bound should be $+p_x\lambda/a$.
- **Lectures 5 and 6, spin slide:** “s = 0, 1, 2, …, n−1” is copied from the ℓ rule. An electron has s = ½ only.
- **Lecture 4:** the slide uses $\Delta y\, \Delta p_y \sim h$ as an order-of-magnitude result, and $\hbar/2$ as the exact bound. Both appear; they are not in conflict.
- **Lecture 6, spin–orbit slide:** “$U = -\mu_S \cdot B_{eff} = L\cdot S$” is shorthand for *proportional to* L·S (the coefficient is $\xi(r)$).
- **Lecture 5/6, Stern–Gerlach slide:** which spot (up or down) belongs to which m_ℓ depends on sign conventions and on the sign of dB/dz. The count of spots is what matters.